

The Lucid Beamer Theme

Every slide shows the code that made it

A copy-and-paste manual

Outline

Getting started

The layout

Text

Figures

Tables, maths, closings

Options and backup

Start a deck: two lines

- ▶ Put `beamerthemelucid.sty` next to your `.tex` file

```
\documentclass[11pt,aspectratio=43]{beamer}  
\usetheme{lucid}
```

- ▶ Use `aspectratio=169` for widescreen

Compile twice. The outline slides need a second pass. `snippets/start.tex`

The title slide

- ▶ **No name, institution or logo appears unless you set one**
- ▶ `[plain]` drops the footer, which a title slide should not have

```
\title{Your Title}  
\subtitle{An optional second line}  
\lucidevent{Venue or occasion}  
\date{}  
% \author{Your Name}           % optional  
% \institute{Your Institution} % optional  
% \lucidlogo{logo.pdf}         % optional  
  
\begin{frame}[plain]  
  \titlepage  
\end{frame}
```

Make a new slide

- ▶ A slide is a **frame**. Title in braces, content between
- ▶ Leave the braces off for a slide with no title

% A slide is a "frame". This is the whole pattern:

```
\begin{frame}{The slide title}
  \begin{itemize}
    \item A point
    \item Another point
  \end{itemize}
\end{frame}
```

% No title? Leave the braces off:

```
\begin{frame}
  \takeaway{A single sentence, no title.}
\end{frame}
```

% No title bar, no footer -- for a full-bleed image or a cover:

```
\begin{frame}[plain]
  \slidefigure[\paperwidth]{cover.pdf}
\end{frame}
```

Make a new slide (cont.)

```
\end{frame}

% Reveal the bullets one at a time:
\begin{frame}{Built up in steps}
  \begin{itemize}
    \item<1-> Appears first
    \item<2-> Appears second
  \end{itemize}
\end{frame}
```

snippets/new-slide.tex

Hide a slide

- ▶ Comment it out to drop it for now
- ▶ Move it below backupframes to keep it without counting it

```
% Wrap anything you want to park. Whole frames are fine.
```

```
\begin{hidden}  
\begin{frame}{Cut for time}  
  \begin{itemize}  
    \item Still here, just not in the deck  
  \end{itemize}  
\end{frame}  
\end{hidden}
```

```
% Then choose a mode in the preamble. Two, and they are opposites:
```

```
\usetheme{lucid}                                % hidden=off is the default:  
                                                % the block is NOT compiled. Not in  
                                                % the PDF, not in the page count.
```

Hide a slide (cont.)

```
\usetheme[hidden=show]{lucid}      % compile them, so you can see what
                                   % you have parked.

% Better than commenting out, which is tedious to undo and invisible to
% a search. Better than deleting, which loses the slide.

% --- Two smaller variants -----

% Keep it in the deck but out of the count -- put it below \backupframes:
\backupframes

\begin{frame}{Held in reserve}
  ...
\end{frame}

% Keep it visible but unnumbered:
\begin{frame}[noframenumbering]{Not counted}
  ...
\end{frame}
```

snippets/hidden-slide.tex

Speaker notes

- ▶ note sits inside the frame it belongs to
- ▶ Nothing is shown unless you ask for it, so notes are safe to leave in

% Write the note inside the frame it belongs to:

```
\begin{frame}{Results}
  \begin{itemize}
    \item The gain holds across both attackers
  \end{itemize}
  \speakernote{Say the number out loud: 78 against 62.
    If asked about variance, point at the backup slide.}
\end{frame}
```

% Then choose a mode in the preamble. Two matter, and they are opposites:

```
\usetheme{lucid}                                % notes=off is the default:
                                                  % notes are NOT compiled at all.
                                                  % This is the file you hand out.
```

Speaker notes (cont.)

```
\usetheme[notes=pages]{lucid}    % each slide is followed by a typeset
                                % notes page, in the deck's own colours.

% Two more, for presenting:
\usetheme[notes=only]{lucid}     % notes without slides -- a script
\usetheme[notes=second]{lucid}   % presenter view, notes on screen two

% Beamer's own \note{...} works too and obeys the same option.
% Tip: build twice -- once plain to present from, once with
% notes=pages to print.
```

snippets/notes.tex

Animation: one slide, several pages

- ▶ Each step prints as **another page** of the PDF

This slide is three pages. In handout mode it collapses back to one.

Animation: one slide, several pages

- ▶ Each step prints as **another page** of the PDF
- ▶ Nothing moves — the reader sees a page at a time

This slide is three pages. In handout mode it collapses back to one.

Animation: one slide, several pages

- ▶ Each step prints as **another page** of the PDF
- ▶ Nothing moves — the reader sees a page at a time
- ▶ So an animated deck has more pages than slides

This slide is three pages. In handout mode it collapses back to one.

Animation: the syntax

```
% An "animation" here is not motion. Each step prints as ANOTHER PAGE
% of the PDF, showing a little more. A frame with three steps becomes
% three pages. That is why an animated deck has more pages than slides.
```

```
% --- Bullets, one at a time -----
```

```
\begin{frame}{Built up}
  \begin{steps}
    \item Appears on page 1
    \item Appears on page 2
    \item Appears on page 3
  \end{steps}
\end{frame}
```

```
% --- Break anywhere -----
```

```
\begin{frame}{Two halves}
  The setup.
  \stepafter
  The punchline.
\end{frame}
```

```
% --- Precise control -----
```

```
\showfrom{2}{Appears from page 2 on; space held before it.}
\showonly{2}{Appears only on page 2; no space held.}
```

Animation: the syntax (cont.)

```
\emphat{2}{Highlighted on page 2, plain otherwise.}
\swap{2}{Before}{After}

% --- Swapping a figure in place -----
\begin{frame}{Same spot, two figures}
  \showonly{1}{\slidefigure{before.pdf}}%
  \showonly{2}{\slidefigure{after.pdf}}%
\end{frame}

% Beamer's own <1->, \pause, \only and \alt still work. Use whichever
% reads better. Keep animation for places where the ORDER matters --
% every step is a page someone has to sit through.

% --- A figure that stays while the text builds -----
% The classic trap: put the overlays INSIDE a pane, not around the
% columns, or the whole split appears only on the last page.
\begin{frame}{Figure holds, text builds}
  \begin{figuretext}[0.55]
    \begin{steps}
      \item First point
      \item Second point
    \end{steps}
  \nextpane
```

Animation: the syntax (cont.)

```
\slidefigure{diagram.pdf}    % no overlay spec: present throughout
\end{figuretext}
\end{frame}
```

```
% Printing an animated deck? See snippets/handout.tex -- one word
% collapses every step back to a single page per slide.
```

snippets/animation.tex

Printing an animated deck

- ▶ **handout** collapses every step back to one page per slide
- ▶ Build three files from one body: the talk, the handout, your script

Printing an animated deck (cont.)

```
% A deck with animation is painful to print: every step is its own page,  
% so a 20-slide talk becomes 60 pages. "handout" flattens that -- each  
% frame collapses back to ONE page with everything shown.
```

```
\documentclass[11pt,aspectratio=43,handout]{beamer}  
\usetheme{lucid}
```

```
% That single word turns off every overlay: \steps, \stepafter,  
% \showfrom, \showonly, \swap and beamer's own <1-> all stop splitting.
```

```
% --- Watch out -----  
% \showonly{n}{...} reserves no space, so two \showonly blocks that  
% shared a spot in the talk land ON TOP of each other in a handout.  
% If a slide swaps one figure for another, use \swap instead:  
\swap{2}{\slidefigure{before.pdf}}{\slidefigure{after.pdf}}
```

```
% --- The three files worth building -----  
% 1. the talk      \documentclass[...]{beamer}  
% 2. the handout   \documentclass[... ,handout]{beamer}  
% 3. your script   \usetheme[notes=only]{lucid}
```

```
% --- Four slides to a page, for printing -----
```

Printing an animated deck (cont.)

```
\usepackage{pgfpages}  
\pgfpagesuselayout{4 on 1}[letterpaper,landscape,border shrink=5mm]  
  
% Keep the talk and the handout as separate .tex files, each a couple of  
% lines long, both \input-ing the same body. Then they cannot drift.
```

snippets/handout.tex

Outline

Getting started

The layout

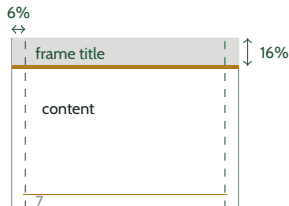
Text

Figures

Tables, maths, closings

Options and backup

The grid, and what you can move



- ▶ Rail, band and footer are **fixed by the theme**
- ▶ You place content inside them; you do not set margins
- ▶ Content starts under the rule and runs down

*% The grid is fixed by the theme. You do not set margins; you place
% content inside them.*

%

% left/right rail 6.0% of page width title and body share it

% title band 16.0% of page height full bleed, grey

% rule under band 0.67% of page height full bleed, amber

% footer rule 92.6% down the page slide number, date

The grid, and what you can move (cont.)

```
%  
% Content starts under the rule and runs down. Useful controls:  
  
% Push content to the bottom of the content area:  
\vfill  
\takeaway{The last thing they should see.}  
  
% Split the slide. The number is the LEFT pane's share:  
\begin{figuretext}[0.55]  
  \begin{itemize}  
    \item Left pane  
  \end{itemize}  
  \nextpane  
  \slidefigure{diagram.pdf}  
\end{figuretext}  
  
% Centre this one frame instead of top-aligning it:  
\begin{frame}[c]{A centred slide}  
  \takeaway{One line, centred.}  
\end{frame}  
  
% Drop the chrome entirely for a full-bleed slide:  
\begin{frame}[plain]
```

The grid, and what you can move (cont.)

```
\slidefigure[\paperwidth]{cover.pdf}  
\end{frame}  
  
% Deck-wide switches:  
% \usetheme[align=center]{lucid}    centre every slide  
% \usetheme[footer=none]{lucid}    keep the rule, drop the text
```

snippets/layout.tex

Outline

Getting started

The layout

Text

Figures

Tables, maths, closings

Options and backup

Bullets: two levels, and a third to avoid

- ▶ Level one
 - ▶ Level two
 - Level three is muted on purpose

```
\begin{itemize}  
  \item Level one  
    \begin{itemize}  
      \item Level two  
    \end{itemize}  
\end{itemize}
```

Aim for two top-level bullets and 25–30 words a slide. snippets/bullets.tex

Emphasis: three devices, and none of them is bold

- ▶ A **primary term**, a *secondary term*, something highlighted
- ▶ Each one survives greyscale printing on its own

```
A \term{primary term}, a \altterm{secondary term},  
something \hl{highlighted}
```

snippets/emphasis.tex

Maths beyond one equation

- Aligned lines, cases, matrices, and numbering when you need it

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_{\pi} \left[\sum_t \gamma^t r_t \right] \\ &= \sum_s d^{\pi}(s) v^{\pi}(s)\end{aligned}$$

$$r(s) = \begin{cases} +1 & \text{host recovered} \\ -1 & \text{host compromised} \\ 0 & \text{otherwise} \end{cases}$$

Maths beyond one equation (cont.)

```
% amsmath is already loaded by the theme. So is a sans maths font, so
% symbols match the body weight instead of sitting in a serif face.

% --- Inline and display -----
The error level  $\lambda$  controls it.

\[ q_j = \frac{e^{\lambda \mathrm{EU}_j(x)}}{\sum_i e^{\lambda \mathrm{EU}_i(x)}} \]

% Display maths is UNNUMBERED by default. A talk has no cross-references.

% --- When you must number one -----
\begin{equation}
\mathcal{L} = \mathbb{E}_{\pi} \left[ \sum_t \gamma^t r_t \right]
\label{eq:objective}
\end{equation}
Referred to later as \eqref{eq:objective}.

% --- Several lines, aligned on = -----
\begin{align*}
\mathcal{L} &= \mathbb{E}_{\pi} \left[ \sum_t \gamma^t r_t \right] \\
&= \sum_s d_{\pi}(s) \, v_{\pi}(s)
\end{align*}
```

Maths beyond one equation (cont.)

```
% align* is unnumbered; align numbers every line.

% --- Cases and matrices -----
\[ r(s) = \begin{cases}
+1 & \text{host recovered} \\
-1 & \text{host compromised} \\
0 & \text{otherwise}
\end{cases} \]

\[ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \]

% --- Pointing at part of an equation -----
% Colour the piece you are talking about:
\[ q_j = \frac{e^{\textcolor{red}{\lambda} \mathbf{E}_j(x)}}{\sum_i e^{\lambda \mathbf{E}_i(x)}} \]

% Or reveal it, then emphasise it, across two pages:
\begin{frame}{Building the objective}
  \[ \mathcal{L} = \textcolor{red}{\textbf{E}}_{\pi} \left[ \sum_t \gamma^t r_t \right] \]
  \showfrom{2}{The expectation is over the policy, not the data.}
\end{frame}
```

Maths beyond one equation (cont.)

```
% --- Statements -----  
\begin{theorem}  
  The equilibrium exists for every finite game.  
\end{theorem}  
  
\begin{definition}  
  A \term{world model} predicts the next observation from the last.  
\end{definition}  
  
% Also available: lemma, corollary, proof, example, block.  
% Give any of them a name: \begin{theorem}[Nash, 1951]  
  
% --- Two habits worth keeping -----  
% \mathrm{} for words inside maths: \mathrm{EU}, not EU  
% \dfrac for a full-size fraction inside inline maths
```

snippets/math.tex

Statements

Definition

A **world model** predicts the next observation from the last.

Theorem (Nash, 1951)

Every finite game has an equilibrium in mixed strategies.

- ▶ **Also:** lemma, corollary, proof, example, block

They take the deck's colours, and their bodies stay upright to read from the back.

Pointing at one part of an equation

$$q_j = \frac{e^{\lambda EU_j(x)}}{\sum_i e^{\lambda EU_i(x)}}$$

- Colour the piece you are talking about, or reveal it with `emphat` and `showfrom`

Colour, highlight, underline

- ▶ a primary term, a secondary term, highlighted, underlined, alerted
- ▶ Or reach into the palette directly: gold, muted, teal

```
% Reach for these first. They are the deck's own vocabulary, so a whole
% talk stays consistent and every one survives greyscale printing.
```

```
\term{a primary term}           % accent, bold
\altterm{a secondary term}      % teal, italic
\hl{highlighted}                % amber wash behind the text
\u{underlined}                  % plain underline
\alert{alerted}                  % beamer's own, mapped to the accent

% --- Arbitrary colour, still from the palette -----
\tint{accent}{text}             % forest green -- structure and emphasis
\tint{gold}{text}               % amber         -- rules and second-level marks
\tint{muted}{text}              % grey-green   -- deprioritised text
\tint{ink}{text}                 % near-black  -- body
\tint{termA}{text}               % forest      -- first defined term
```

Colour, highlight, underline (cont.)

```
\tint{termB}{text}      % teal          -- second defined term

% --- Raw colour, if you really need it -----
{\color{lucidAccent} some words}
\textcolor{lucidGold}{some words}

% Palette names you can use anywhere, including in tikz and tables:
%   lucidInk lucidAccent lucidGold lucidMuted lucidBand lucidWash
%   lucidTermA lucidTermB

% --- Colouring part of a table or figure -----
Ours & \term{78} \\
\rowcolor{lucidBand} header & row \\
\fill[lucidAccent] (0,0) rectangle (1,2);    % inside tikz

% Restraint is the point: the reference decks use three or four colours
% for ~95% of all text. Bold is about 3%. If everything is emphasised,
% nothing is.
```

snippets/color.tex Three or four colours carry 95% of the text in a good deck.

Outline

Getting started

The layout

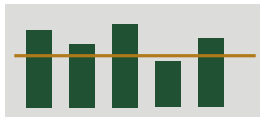
Text

Figures

Tables, maths, closings

Options and backup

A figure, uncaptioned



```
\slidefigure{plot.pdf}  
\slidefigure[0.9\textwidth]{plot.pdf}    % wider or narrower  
\figcaption{Only if a caption is truly needed.}
```

snippets/figure.tex

A figure beside text

- ▶ The left pane holds your points
- ▶ The argument sets its share



```
\begin{figuretext}[0.55]
  \begin{itemize}
    \item Your points
  \end{itemize}
  \nextpane
  \slidefigure{diagram.pdf}
\end{figuretext}
```

snippets/figure-text.tex

Outline

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Options and backup

A table

Method	Score
Baseline	62
Ours	78

```
\begin{lucltable}  
\begin{tabular}{lr}  
  \toprule  
  \thead{Method & Score} \\  
  \midrule  
  Baseline & 62 \\  
  Ours      & \term{78} \\  
  \bottomrule  
\end{tabular}  
\end{lucltable}
```

Horizontal rules only. snippets/table.tex

An equation

- Equations are unnumbered, and sit inside the bullet that introduces them

$$q_j = \frac{e^{\lambda \text{EU}_j(x)}}{\sum_i e^{\lambda \text{EU}_i(x)}}$$

```
\begin{itemize}
\item Probability of attacking target  $j$ 
  \[ q_j = \frac{e^{\lambda \text{EU}_j(x)}}{\sum_i e^{\lambda \text{EU}_i(x)}} \]
\end{itemize}
```


A source line, and a closing statement

The one sentence you want them to leave
with.

```
\takeaway{The one sentence you want them to leave with.}  
\citeline{Author, A. (2020). Title. Venue.}
```

A source line sits just above the footer rule, like this one.

Code on a slide

- ▶ A frame holding verbatim text must be [fragile]

```
for step in range(n_steps):  
    loss = model.update(batch)
```

% In the preamble, once:

```
\usepackage{listings}  
\lstset{  
    basicstyle=\ttfamily\footnotesize\color{luclidInk},  
    keywordstyle=\color{luclidAccent},  
    commentstyle=\color{luclidMuted}\itshape,  
    backgroundcolor=\color{luclidBand!40},  
    breaklines=true, columns=fullflexible,  
    xleftmargin=5pt, frame=none, showstringspaces=false,  
}
```

% On a slide. The [fragile] is required for verbatim text:

```
\begin{frame}[fragile]{Training loop}  
\begin{lstlisting}[language=Python]  
for step in range(n_steps):  
    loss = model.update(batch)
```

Code on a slide (cont.)

```
\end{lstlisting}  
\end{frame}
```

% Long or awkward code is safer in its own file:

```
\lstinputlisting[language=Python]{code/train.py}
```

% Trap: a [fragile] frame ends at the first \end{frame} at the start of

% a line -- including one that is only sample text inside a listing.

% If you must show \end{frame}, use \lstinputlisting from a file.

snippets/code.tex

Citing sources

- ▶ citeline is the light option: a credit line, no .bib needed
- ▶ For real citations, use a .bib and cite as usual [1]

```
% --- 1. a light source line, no .bib file needed -----
\citeline{McKelvey \& Palfrey (1995). Games and Economic Behavior.}

% --- 2. real citations from a .bib file -----
% Cite in the text:
Quantal response \cite{mckelvey1995quantal} explains it.

% Then one frame at the end of the deck:
\begin{frame}[allowframebreaks]{References}
  \footnotesize
  \bibliographystyle{plain}
  \bibliography{refs}
\end{frame}

% Build order: pdflatex, bibtex, pdflatex, pdflatex.
```

Citing sources (cont.)

```
% Overleaf and latexmk do this for you.
```

Build order: pdflatex, bibtex, pdflatex, pdflatex.

Sections make their own divider

- ▶ Every `section` inserts an outline slide with the current section marked
- ▶ You have passed four of them already

```
\section{Your section name}  
  
% To place the outline slide by hand instead:  
% \settheme[outline=manual]{lucid}  
% \outlineframe
```

snippets/sections.tex

Outline

Getting started

The layout

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Options and backup

Theme options

Option	Values, default first
palette	green, maroon, slate
font	cabin, helvet, none
footer	full, page, none
pagenumber	plain, total
outline	auto, manual
align	top, center
blind	false, true

```
\usetheme[palette=maroon,pagenumber=total]{lucid}
```


Anonymous submission

- ▶ **Nothing identifying is emitted unless you set it**
- ▶ `blind` suppresses author, institution and logo even when set, so a deck flips to anonymous without editing its front matter

```
\usetheme[blind]{lucid}
```

snippets/blind.tex

Backup slides

- ▶ `backupframes` stops later slides inflating the page count

```
\backupframes  
  
\begin{frame}{Backup: extra results}  
  ...  
\end{frame}
```

The next slide is a backup slide. Watch the page number restart.

References



Richard D. McKelvey and Thomas R. Palfrey.
Quantal response equilibria for normal form games.
Games and Economic Behavior, 10(1):6–38, 1995.

Backup: the numbering restarted

- ▶ This is a backup slide
- ▶ The footer count started again at 1